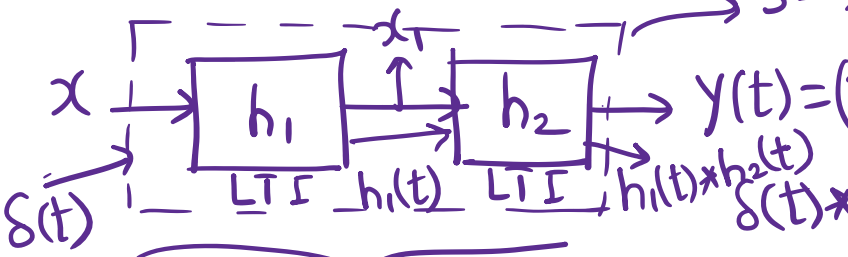
$$x(t) \longrightarrow \boxed{h(t)} \longrightarrow y(t) = x(t) * h(t)$$

LTI

$$= \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$

$$(x * h_1) * h_2 = x * (h_1 * h_2)$$

$S \rightarrow h$



$$y(t) = (x(t) * h_1(t)) * h_2(t) \quad -①$$

$$\delta(t) * h_1(t) = \int_{-\infty}^{\infty} \delta(\tau)\, h_1(t-\tau)\, d\tau$$

$$= \int_{-\infty}^{\infty} h_1(\tau)\, \delta(t-\tau)\, d\tau$$

$$= h_1(t)$$

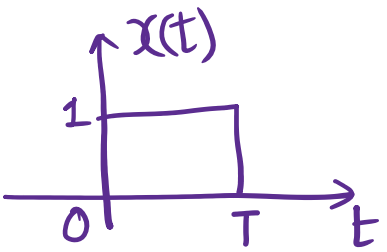
$$h(t) = h_1(t) * h_2(t)$$

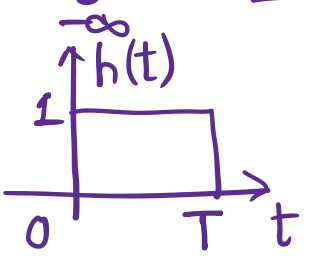
$$y(t) = x(t) * h(t)$$

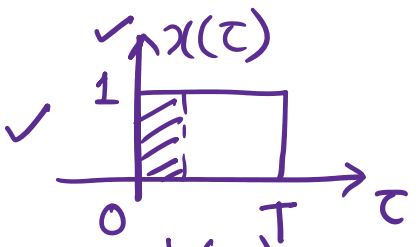
$$= x(t) * (h_1(t) * h_2(t)) \quad -②$$

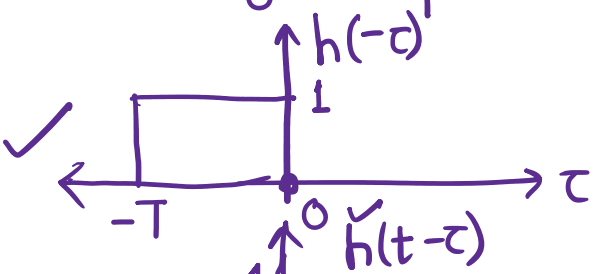
$\therefore$ ① & ② $\Rightarrow (x * h_1) * h_2 = x * (h_1 * h_2)$

Ex. 1) . $y(t) \triangleq x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$



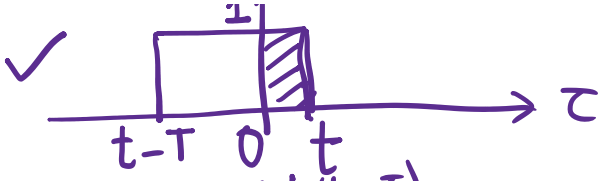


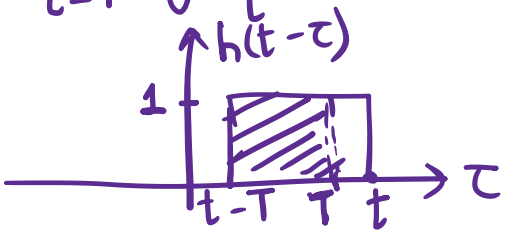




$$y(t) = \begin{cases} 0 & , t < 0 \\ 0 & , t > 2T \end{cases}$$

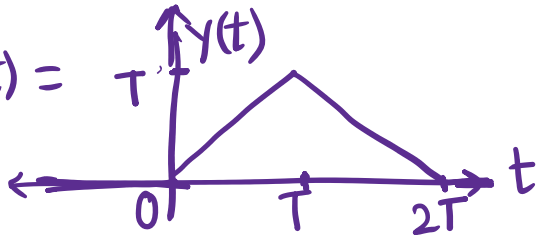
$, \; 0 \le t \le T \Rightarrow y(t) = t$



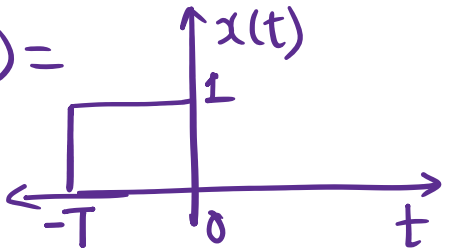


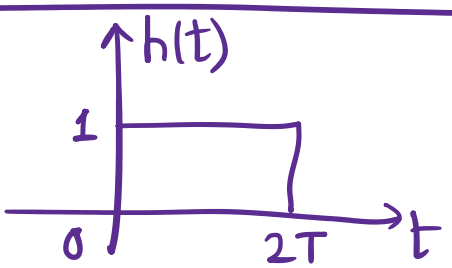
$T \le t \le 2T \Rightarrow y(t) = T-(t-T)$
$= 2T - t$

$y(t) =$

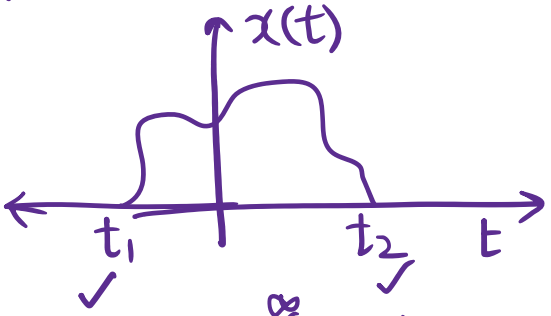


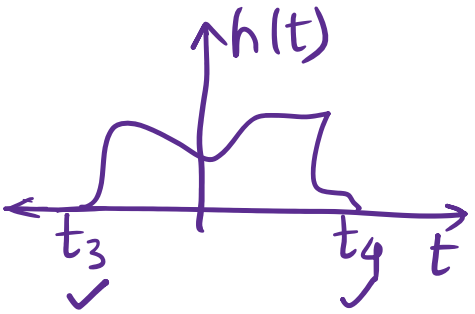
Ex. 2) $x(t) =$





$y(t) \triangleq x(t) * h(t)$?





$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau$$

$x(\tau)$: $t_1 \le \tau \le t_2$

$h(t-\tau)$: $t_3 \le t-\tau \le t_4$

$t_1 + t_3 \le t \le t_2 + t_4$

$y(t)$ can be non-zero ONLY bet^n $\underbrace{[t_1+t_3,\ t_2+t_4]}_{\text{Support of } y(t)}$

$x(t) \rightarrow \boxed{h(t)} \rightarrow y(t) = x(t) * h(t)$

✓ Memoryless?

✓ Causal?

LTI ✓ Stable?

① Memoryless / With memory : (LTI)

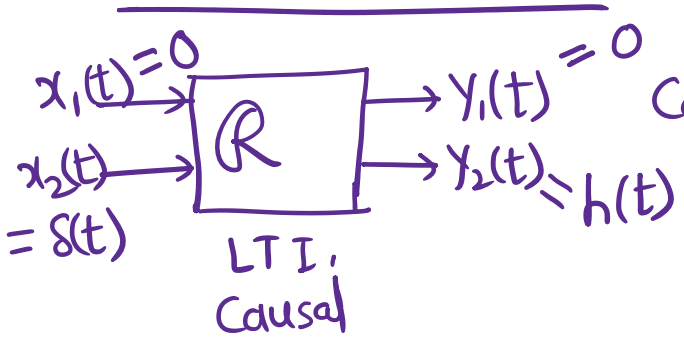
$y(t) = f(x(t))$, $x^2(t)$, $kx(t)$, $|x(t)|$

$y(t) = \alpha x(t)$, $\alpha \in \mathbb{C}$

$\Rightarrow h(t) = \alpha \delta(t)$

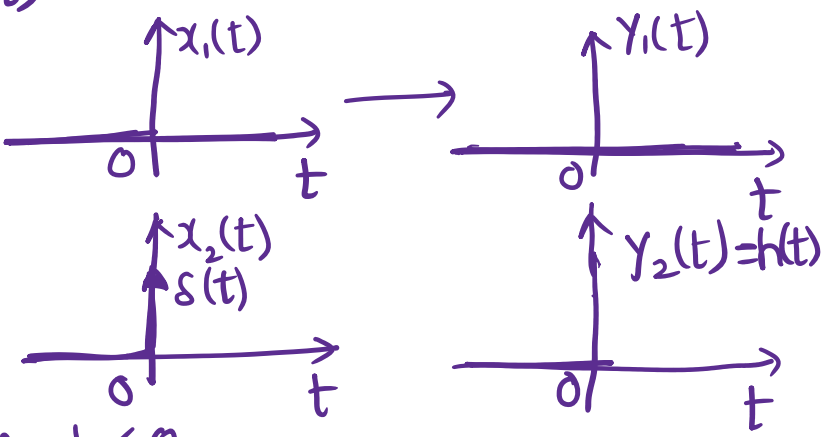
$\therefore$ Memoryless LTI $\Longleftrightarrow h(t) = \alpha \delta(t)$

② Causal / N.C. LTI :



Causal $\Leftrightarrow x_1(t) = x_2(t)$, $t \le t_0$

$\Rightarrow y_1(t) = y_2(t)$, $t \le t_0$



$x_1(t) = x_2(t)$, $t < 0$

$\Rightarrow y_1(t) = y_2(t)$, $t < 0$

$\Rightarrow 0 = h(t)$, $t < 0$

i.e. Causal LTI $\Rightarrow h(t) = 0$, $t < 0$

Let $h(t) = 0$, $t < 0$

$$y(t) = \int_{-\infty}^{\infty} h(\tau) x(t-\tau)\, d\tau = \int_{0}^{\infty} h(\tau) x(t-\tau)\, d\tau$$

depends on present & past values of i/p $\therefore$ Causal

$\therefore$ Causal LTI $\Longleftrightarrow h(t) = 0$, $t < 0$

③ BIBO stable / Unstable LTI

(3) BIBO stable / unstable LTI

$$y(t) = \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau \qquad x(t) \longrightarrow \boxed{h(t)} \longrightarrow y(t)$$

Let $|x(t)| \le B_x < \infty$

$$|y(t)| = \left| \int_{-\infty}^{\infty} h(\tau) x(t-\tau)\, d\tau \right| \le \int_{-\infty}^{\infty} |h(\tau)|\, \underline{|x(t-\tau)|}\, d\tau$$

$$\le \underbrace{B_x \underbrace{\int_{-\infty}^{\infty} |h(\tau)|\, d\tau}_{<\infty}}_{\le B_y < \infty \ (B_x \text{ finite})}$$

$$\int_{-\infty}^{\infty} |h(t)|\, dt < \infty \Rightarrow \text{BIBO stable}$$

$$y(t) = \int_{-\infty}^{\infty} h(\tau) x(t-\tau)\, d\tau \quad , \ \underline{\text{LTI \& stable}}$$

$$y(0) = \int_{-\infty}^{\infty} h(\tau) x(-\tau)\, d\tau \Rightarrow |y(0)| < \infty \text{ for bounded } x(t)$$

Let $x(-t) \triangleq \operatorname{sgn}(h(t)) = \begin{cases} 1 & \text{if } h(t) > 0 \\ -1 & \text{if } h(t) < 0 \end{cases} \Rightarrow |x(t)| = 1 \ \forall t \Rightarrow x(t) \text{ is bounded}$

$$\Rightarrow y(0) = \int_{-\infty}^{\infty} |h(\tau) \operatorname{sgn}(h(\tau))\, d\tau = \int_{-\infty}^{\infty} |h(\tau)|\, d\tau < \infty$$

$$\Rightarrow \text{Stable, LTI} \Leftrightarrow \int_{-\infty}^{\infty} |h(\tau)|\, d\tau < \infty$$